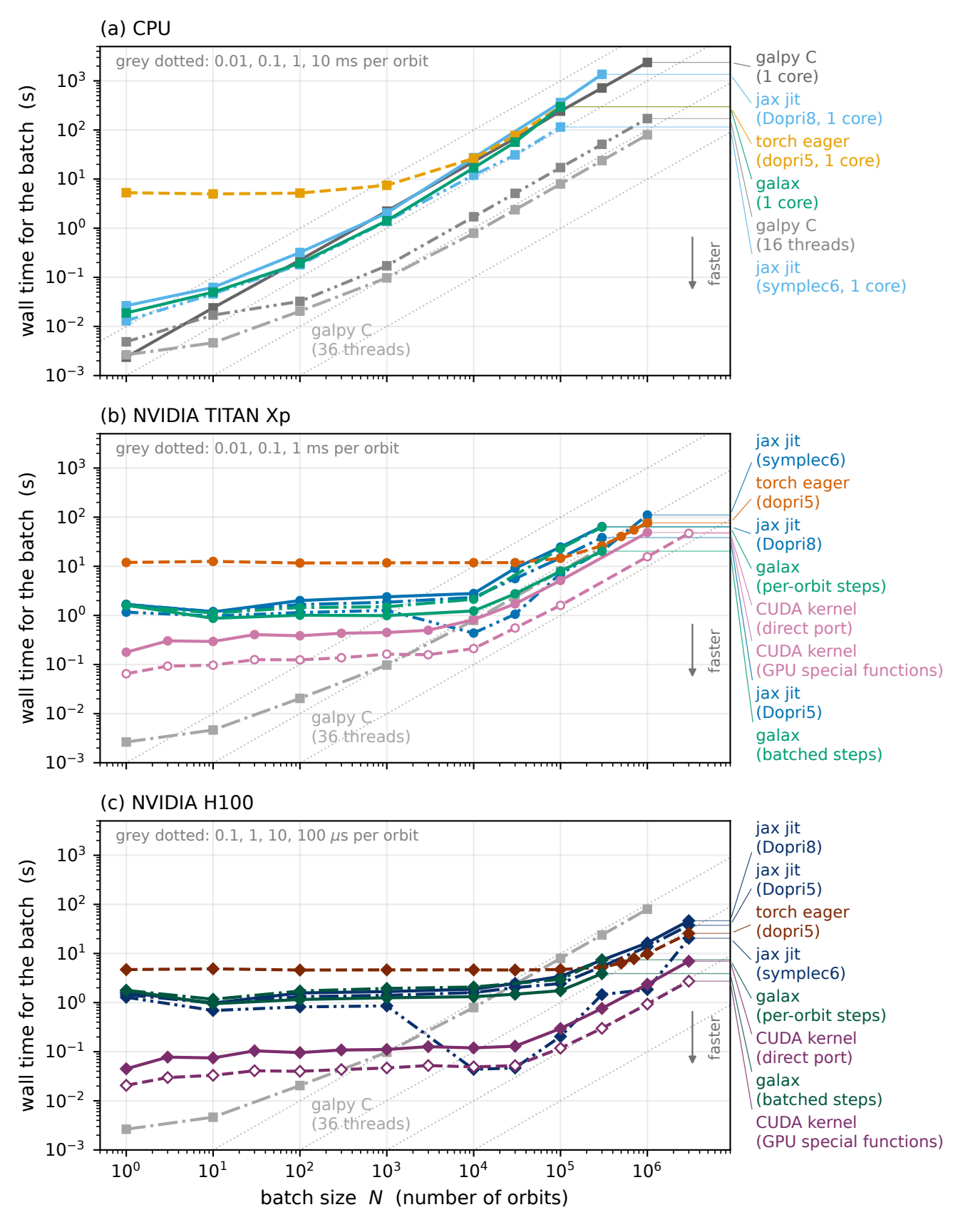
(a) CPU
grey dotted: 0.01, 0.1, 1, 10 ms per orbit
wall time for the batch (s)
galpy C
(36 threads)
faster
galpy C
(1 core)
jax jit
(Dopri8, 1 core)
torch eager
(dopri5, 1 core)
galax
(1 core)
galpy C
(16 threads)
jax jit
(symplec6, 1 core)
(b) NVIDIA TITAN Xp
grey dotted: 0.01, 0.1, 1 ms per orbit
wall time for the batch (s)
galpy C
(36 threads)
faster
jax jit
(symplec6)
torch eager
(dopri5)
jax jit
(Dopri8)
galax
(per-orbit steps)
CUDA kernel
(direct port)
CUDA kernel
(GPU special functions)
jax jit
(Dopri5)
galax
(batched steps)
(c) NVIDIA H100
grey dotted: 0.1, 1, 10, 100 μs per orbit
wall time for the batch (s)
galpy C
(36 threads)
faster
jax jit
(Dopri8)
jax jit
(Dopri5)
torch eager
(dopri5)
jax jit
(symplec6)
galax
(per-orbit steps)
CUDA kernel
(direct port)
galax
(batched steps)
CUDA kernel
(GPU special functions)
batch size N (number of orbits)